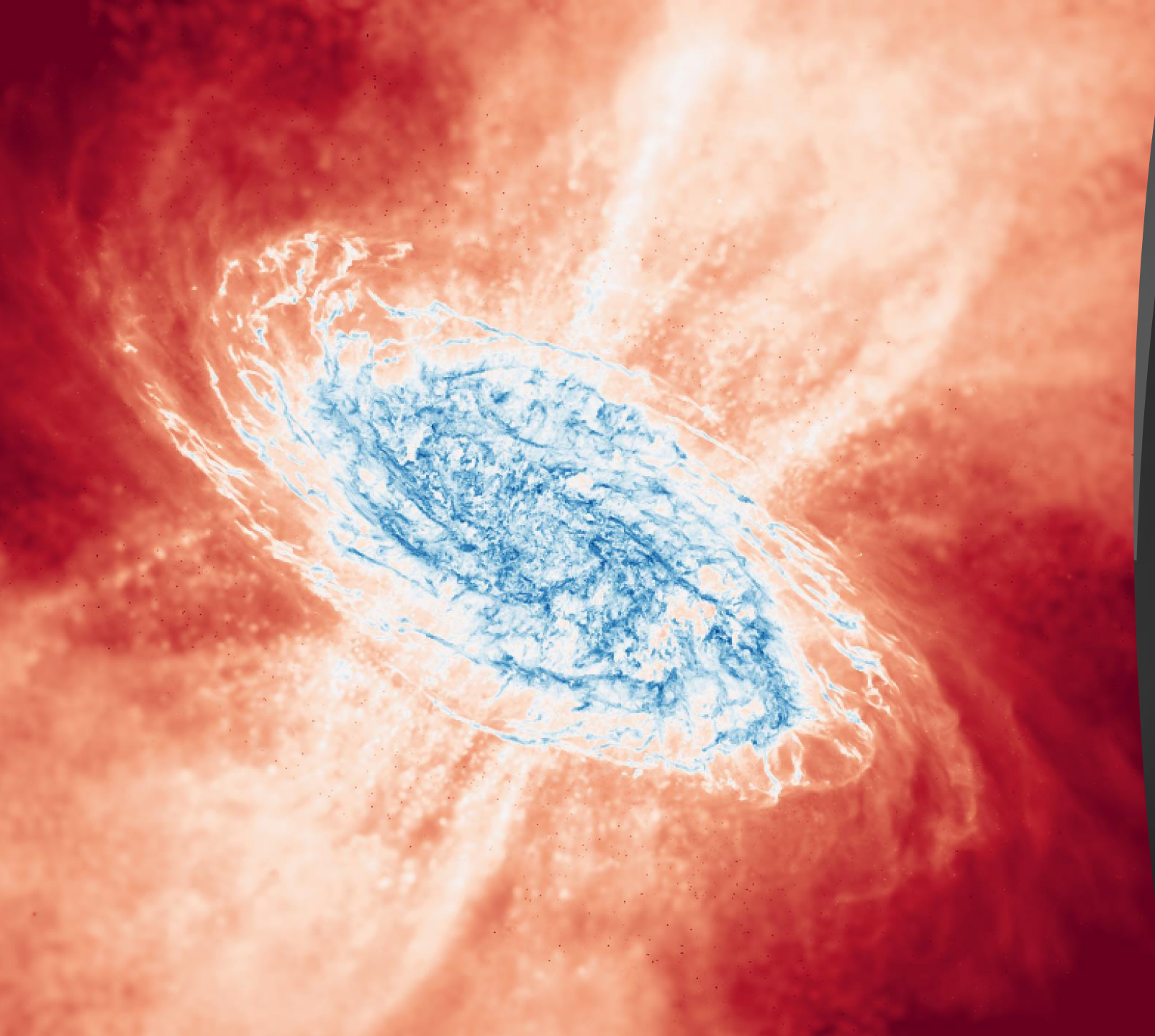


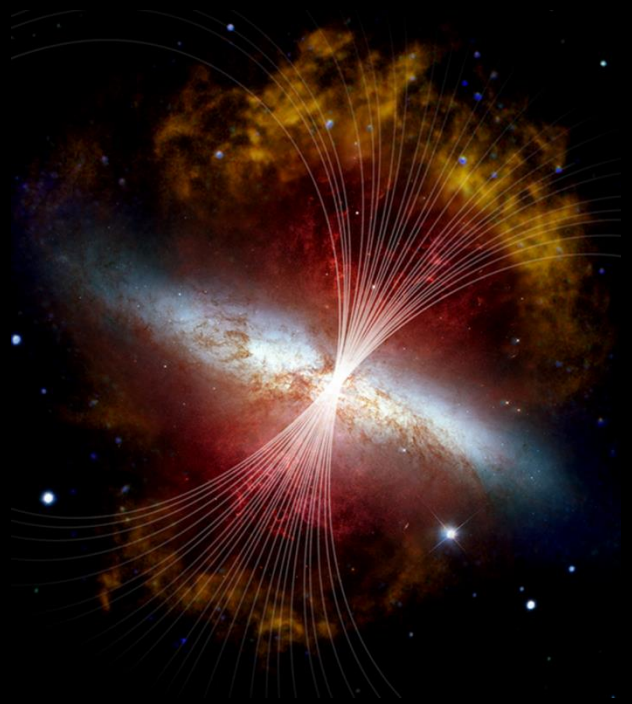
Recent Improvements in SPMHD modeling of galaxies

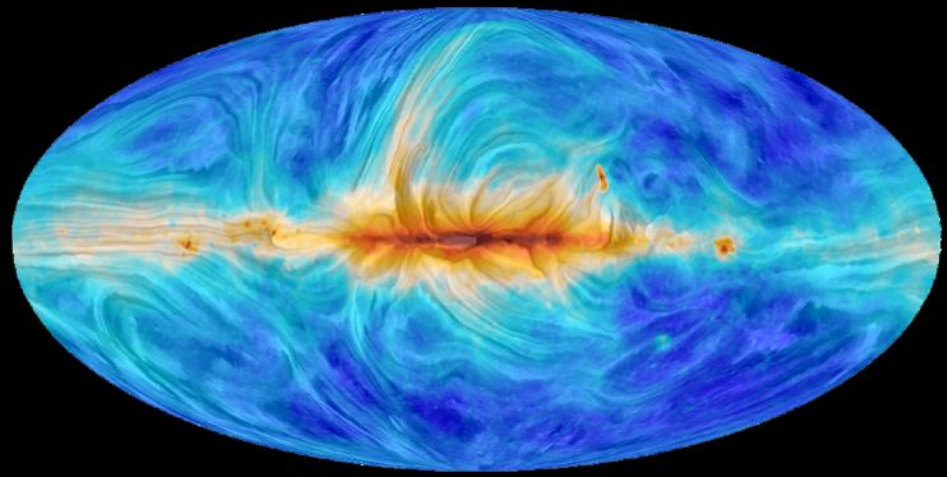
Robert Wissing

Institute of Theoretical Astrophysics, University of Oslo

Magnetic fields in galaxies







Planck Satellite
Milky Way (Planck Collaboration 2016)

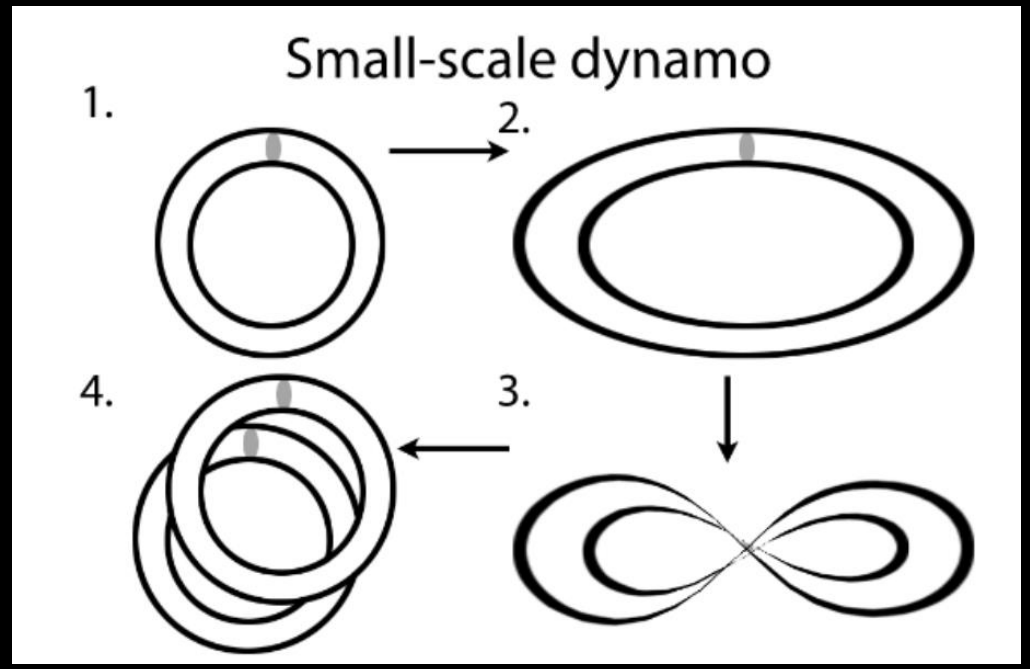
- Ubiqitious in our Universe: the Earth, the Sun, in star forming region (GMCs), galaxies and the CGM/IGM;
- Essential component in star formation;
- Can reach equipartition with thermal and turbulent energy in galaxies -> dynamically important (e.g., Beck+2015);
- Transport of cosmic rays, potentially driving galactic outflows (e.g., Girichidis+2016a);

Origins and amplification of the B-fields

- Primordial seed B fields are *weak*: *e.g.* 10^{-20} G from Bierman battery
- Stellar winds, SNe, AGN feedback inject additional B fields;
- Observations -> ~ μG in the MW- a few 100 μG in starburst galaxies (e.g., M82)
- B-fields are amplified already in high-z galaxies;
  Strong regular fields of 10 μG at z ~ 2 (Bernet et al. 2008; 2013; Kronberg et al. 2008; Farnes et al. 2014b; 2017)
  Few μG fluctating fields at z ~ 4-5 *(Kim et al. (2016))*
- Galactic dynamo with exponential growth required;
  - Small-scale dynamo (At scale < turbulence driving)
  - Mean-field dynamo (At scale > turbulence driving)

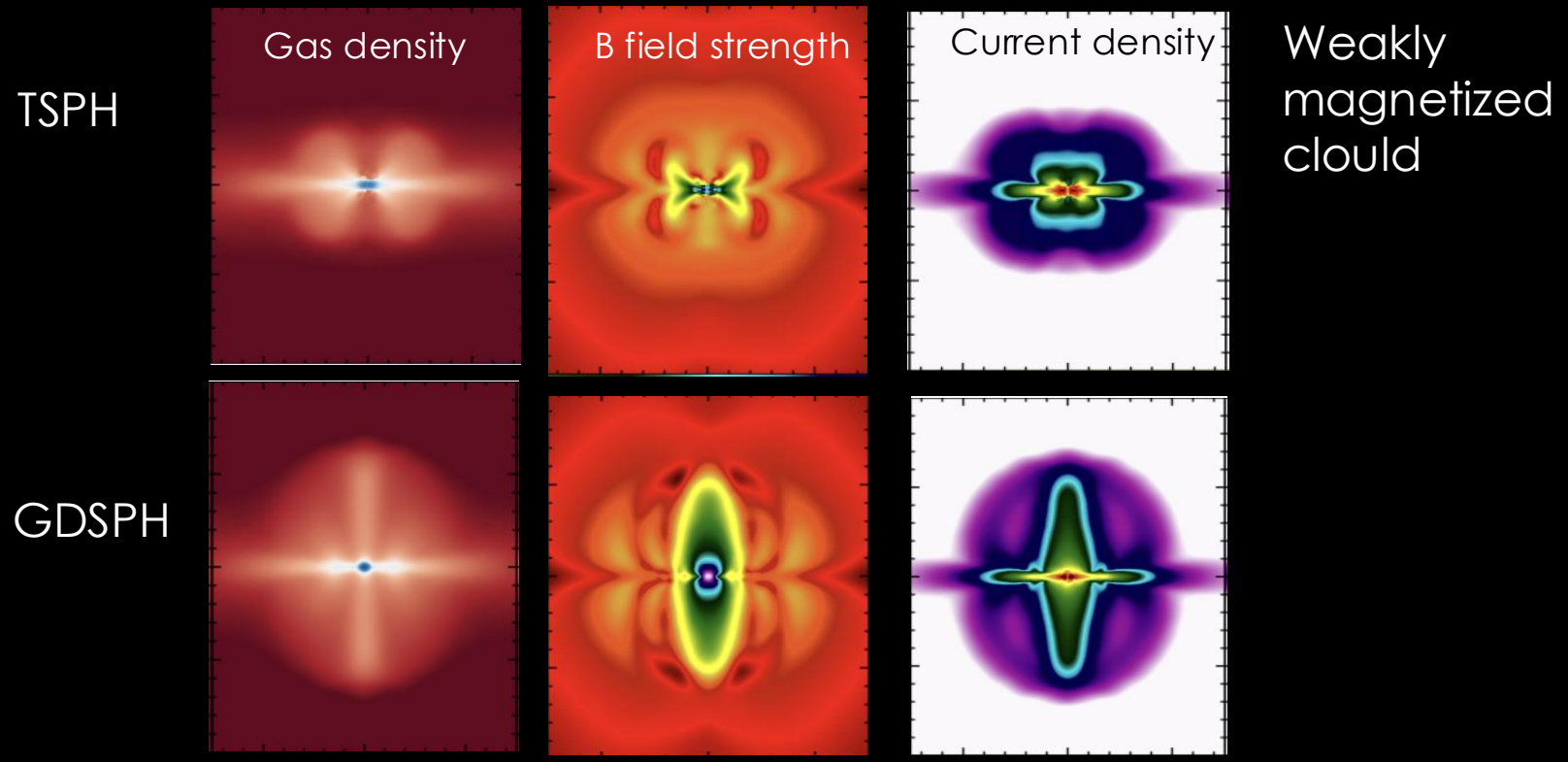
Turbulent dynamos

- Small-scale dynamo
  - Streching, twisting and folding of field lines by turbulent flows
  - Create randomly oriented B-field
  - e-folding time: viscous scale eddies (smaller scale amplies faster); 10-100 Myr in galaxies (Schekochihin+2004)
  - **Solenoidal** (rotational) mode turbulence is essential to excite the dynamo;
- Mean-field dynamo
  - Generate large-scale, ordered B field
  - $\alpha\omega$ dynamo (Parker+1955): shearing motion (ω-effect) and helicities of the velocity fields (α-effect);
  - Other mean-field dynamos also relies on turbulence interacting with large scale density/flow structure



New SPMHD implementation

- Implemented within ChaNGa/Gasoline
- Employs new gradient operators (GDSPH)
- Jet formation captured at much lower resolution and for weaker magnetic fields
- Similar results to MFM/MFV in Gizmo in cloud collapsing/jet launching (Hopkins & Ravis 2016)

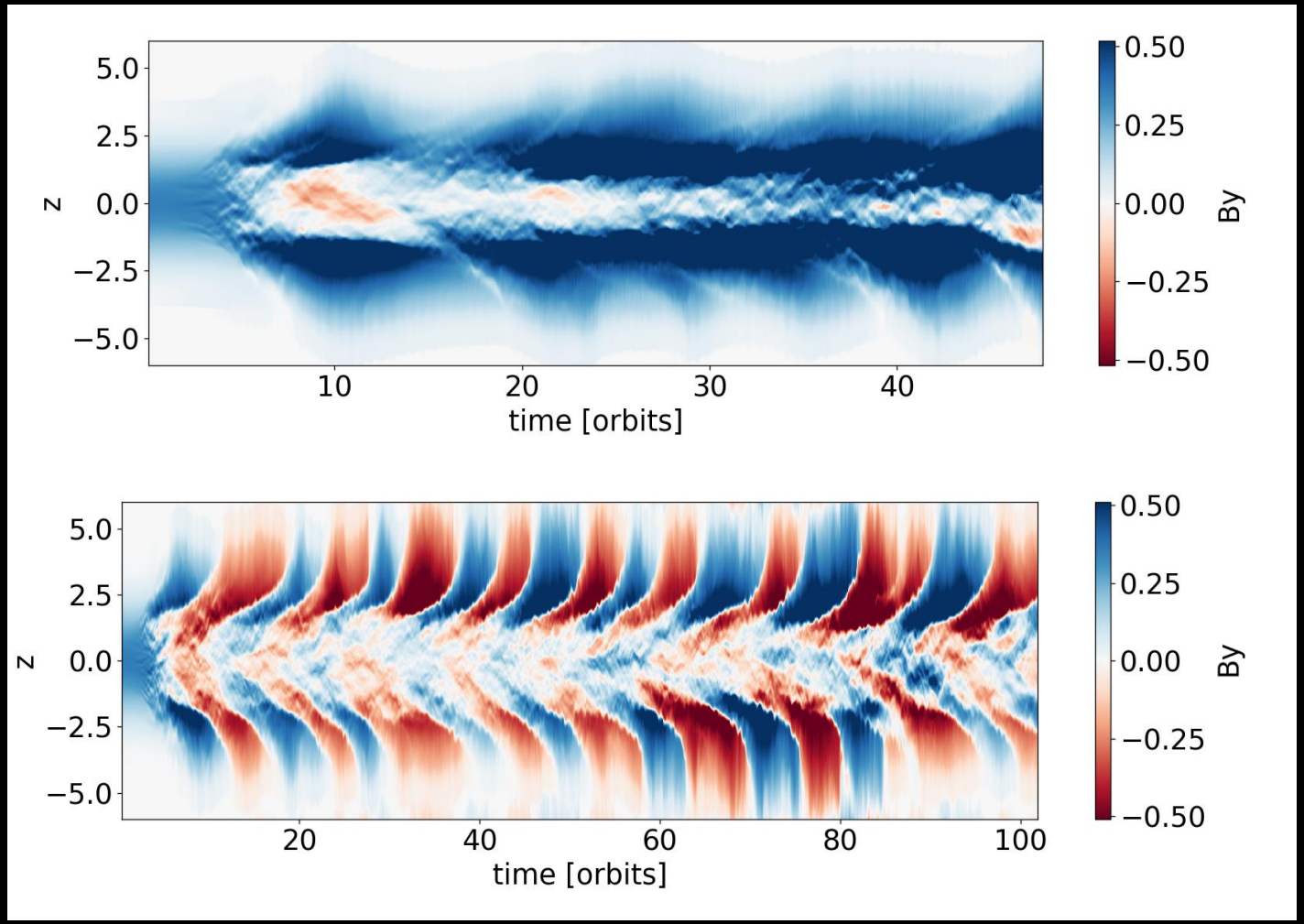


Wissing & Shen (2020)

Magneto rotational instability in stratified medium

TSPH

GDSPH



Wissing et al. (2022)

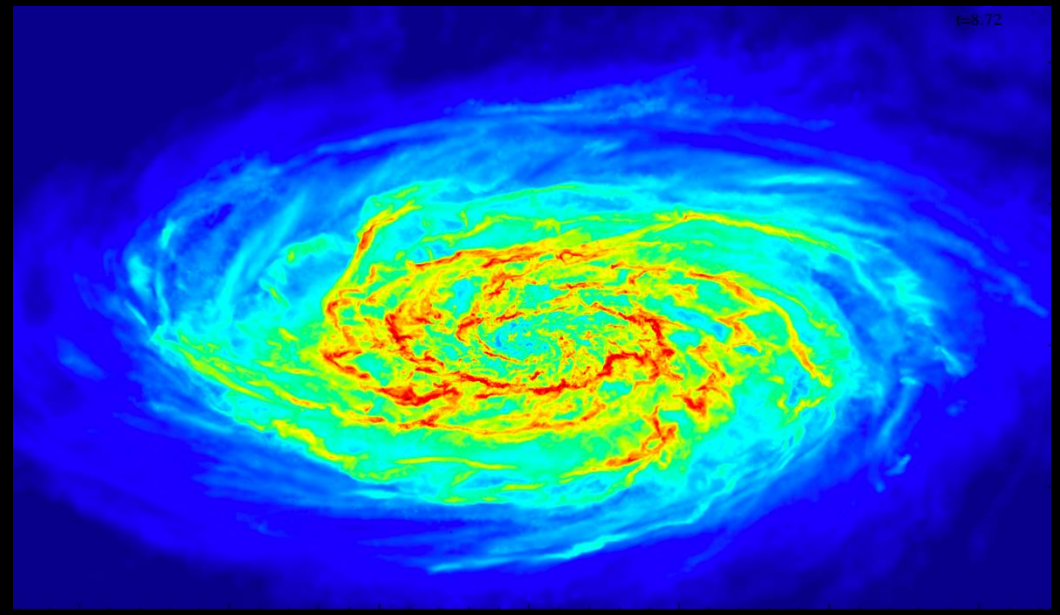
Galaxy Simulations

- 53 Isolated disk from AGORA comparison project, MW-like disks (Kim+2015);
- UVB (z = 0), radiative heating/cooling; star formation, SN feedback;
- Parameter space:
  - Resolution (low, medium, high)
  - Jeans floor (minimum collapsing length);
  - Feedback models:
    - delayed cooling (Stinson+2006) vs. superbubble (Keller+2014);
    - Different injection length;
    - Feedback strength
  - Diffusion parameters;
  - Initial B field configuration;

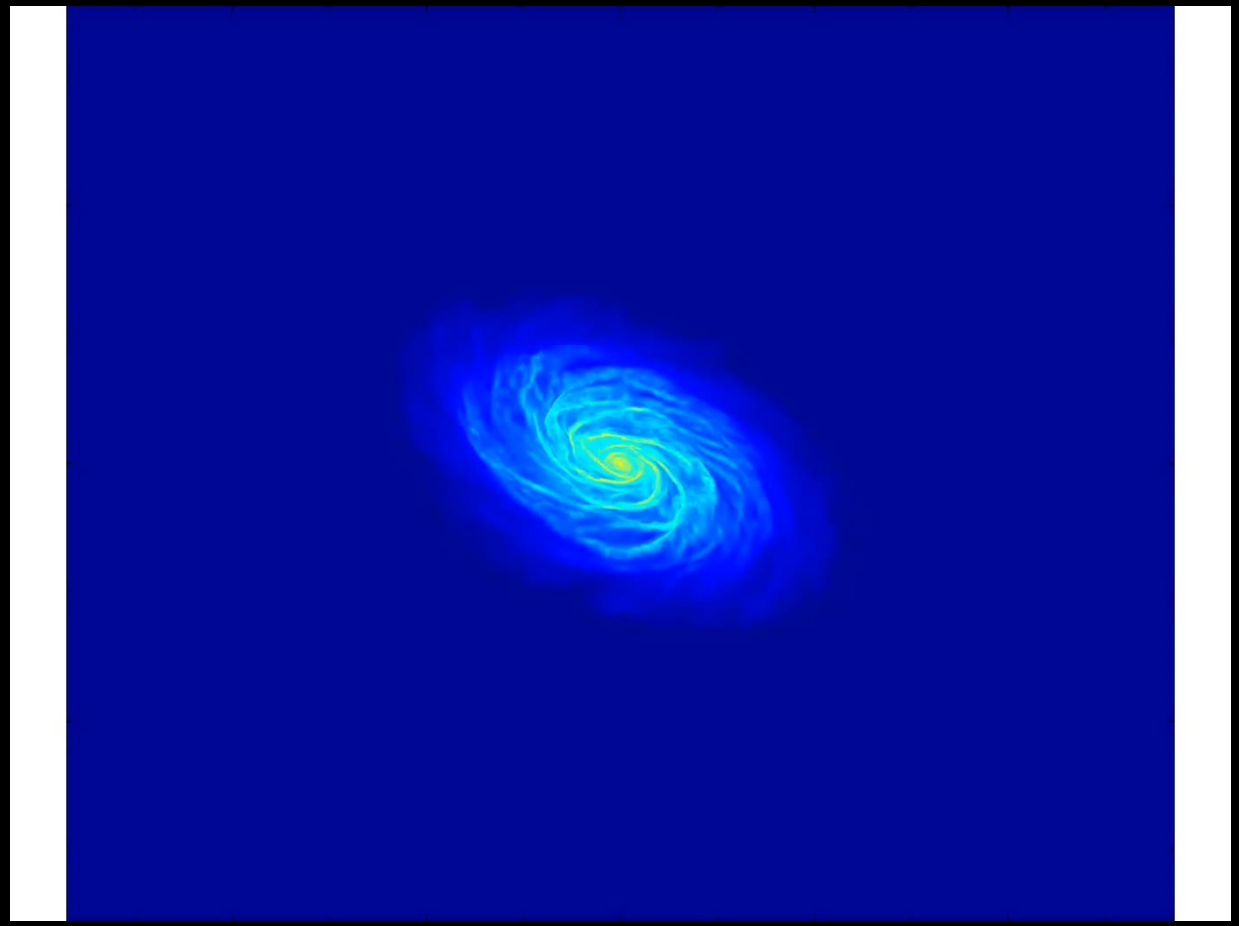
Wissing & Shen (2023)

	N_{tot}	$m_{gas}[M_\odot]$	$m_{star}[M_\odot]$	$m_{dark}[M_\odot]$	$\epsilon_{gas}[kpc]$
Low	$2.5 \cdot 10^6$	$1.0 \cdot 10^4$	$4.3 \cdot 10^4$	$1.6 \cdot 10^6$	0.02
Medium	$2.0 \cdot 10^7$	$1.3 \cdot 10^3$	$5.4 \cdot 10^3$	$2.0 \cdot 10^5$	0.01
High	$1.6 \cdot 10^8$	$1.7 \cdot 10^2$	$6.7 \cdot 10^2$	$2.5 \cdot 10^4$	0.005

¼ of AGORA cosmological zoom-in (Roca-Fabrega+2024)



B-field amplified by feedback

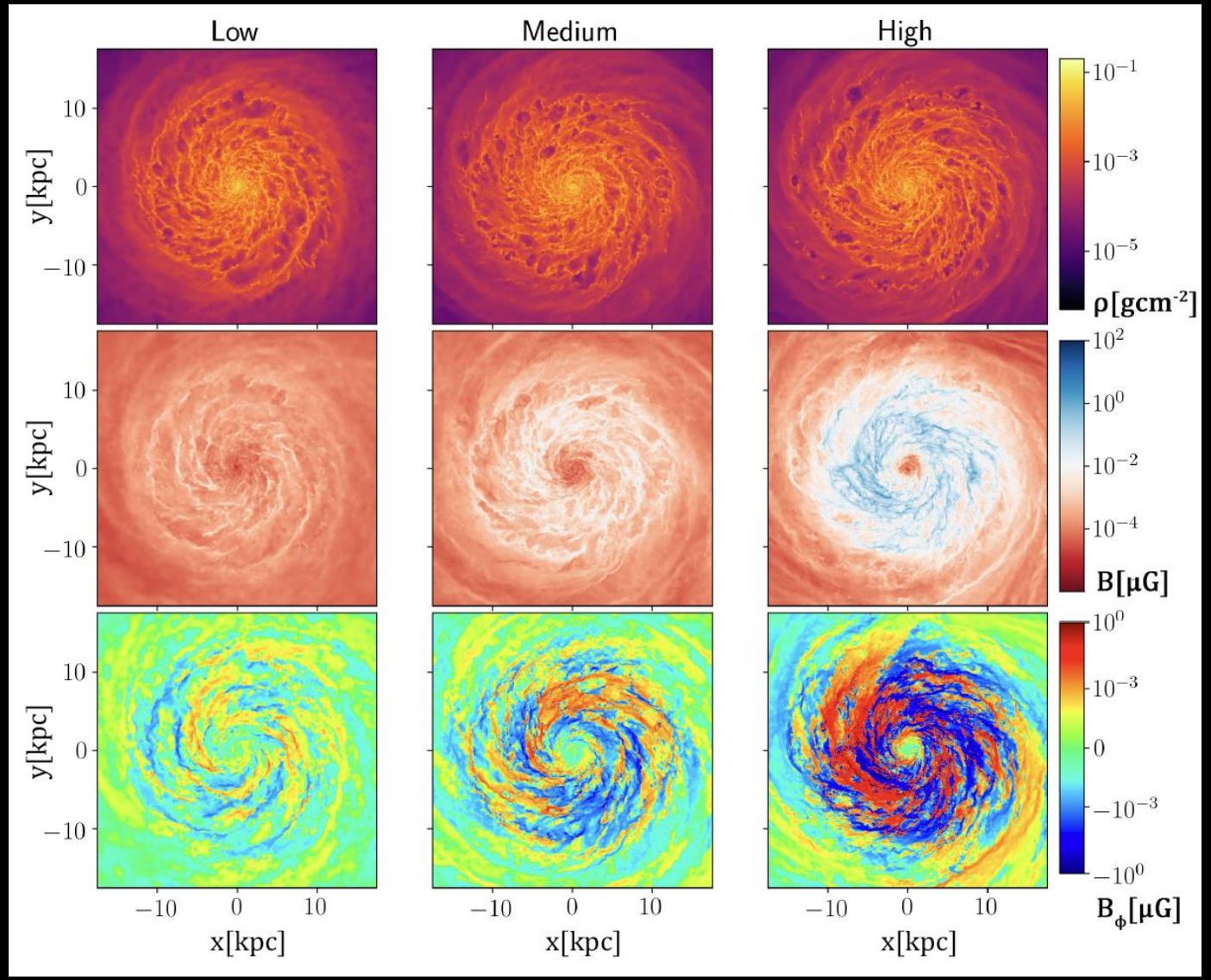


- Superbubble model, with N_{FB} = 1, E_{SN} = 10^{51} ergs, medium resolution (reaches saturation ~ 1.2 Gyr)
- Feedback
  - Advect/diffuse magnetic field, drag into the CGM (destructive esp. central region)
  - Outflow/fountain increase vetical motion -> increase $\alpha\omega$ dynamo
- Magnetization of the CGM

Resolution dependence

Early time t = 250 Myr

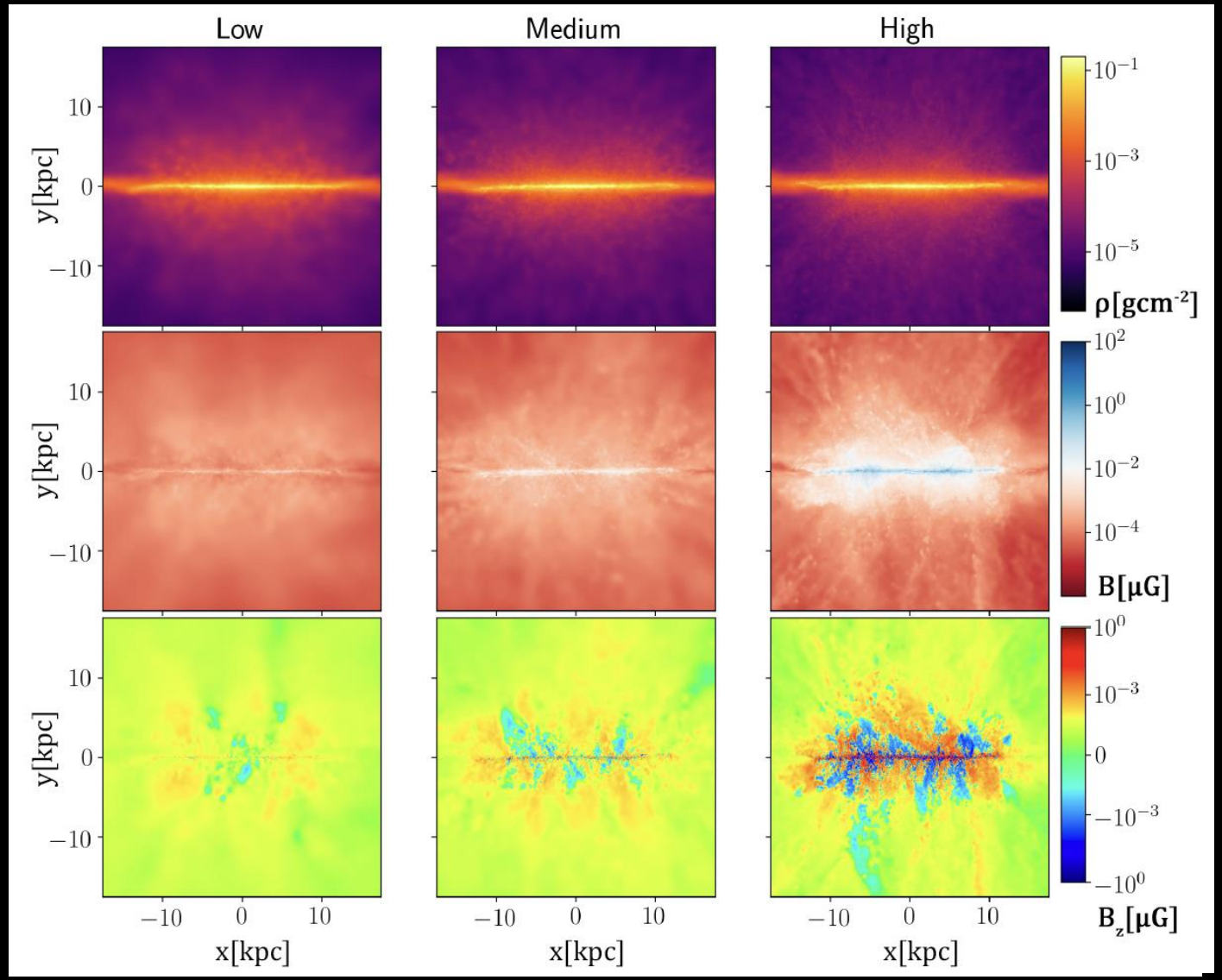
- Superbubble model, with $N_{FB} = 1$, $E_{SN} = 10^{51}$ ergs
- Higher resolution -> faster amplification (not surprising…)
- Polarities and large-scale coherent B-field in all cases-> mean field dynamo
- Complexities in the CGM (near disc plane) shown in higher resolution



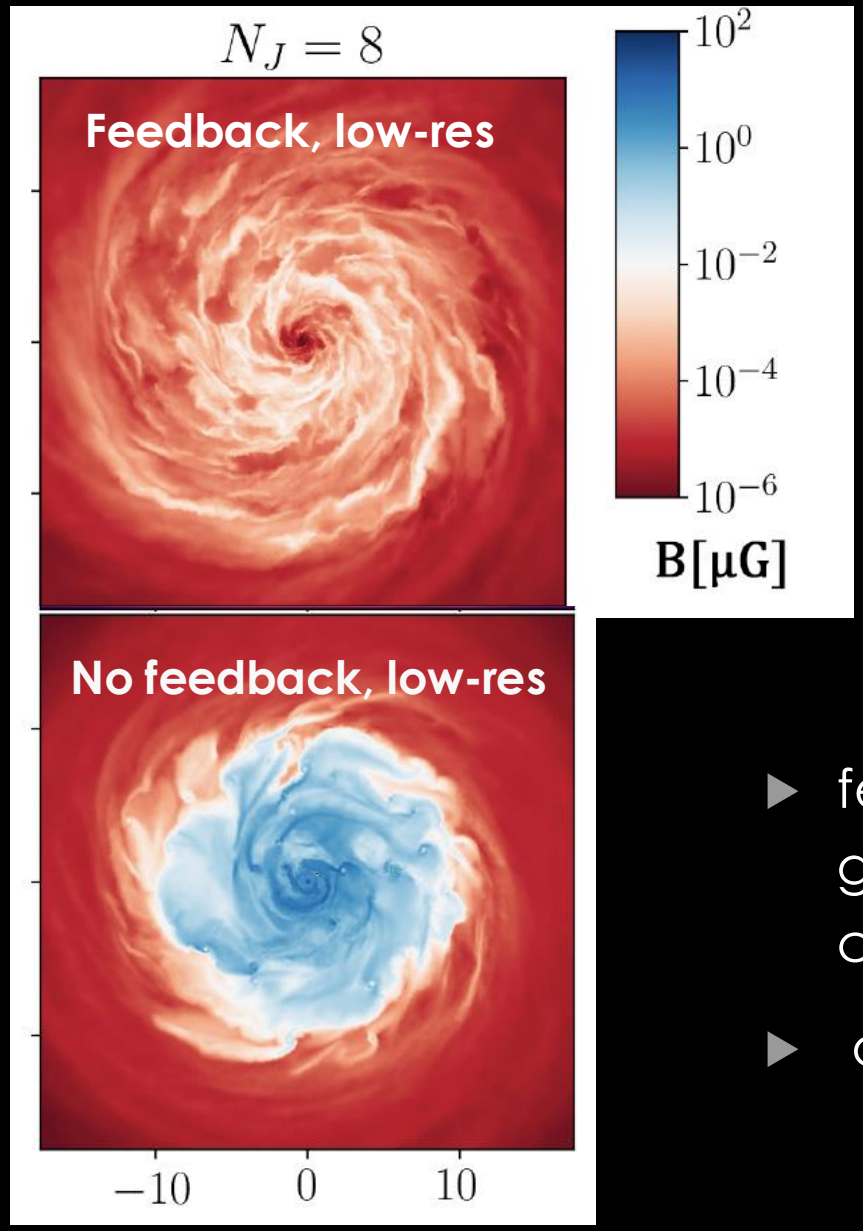
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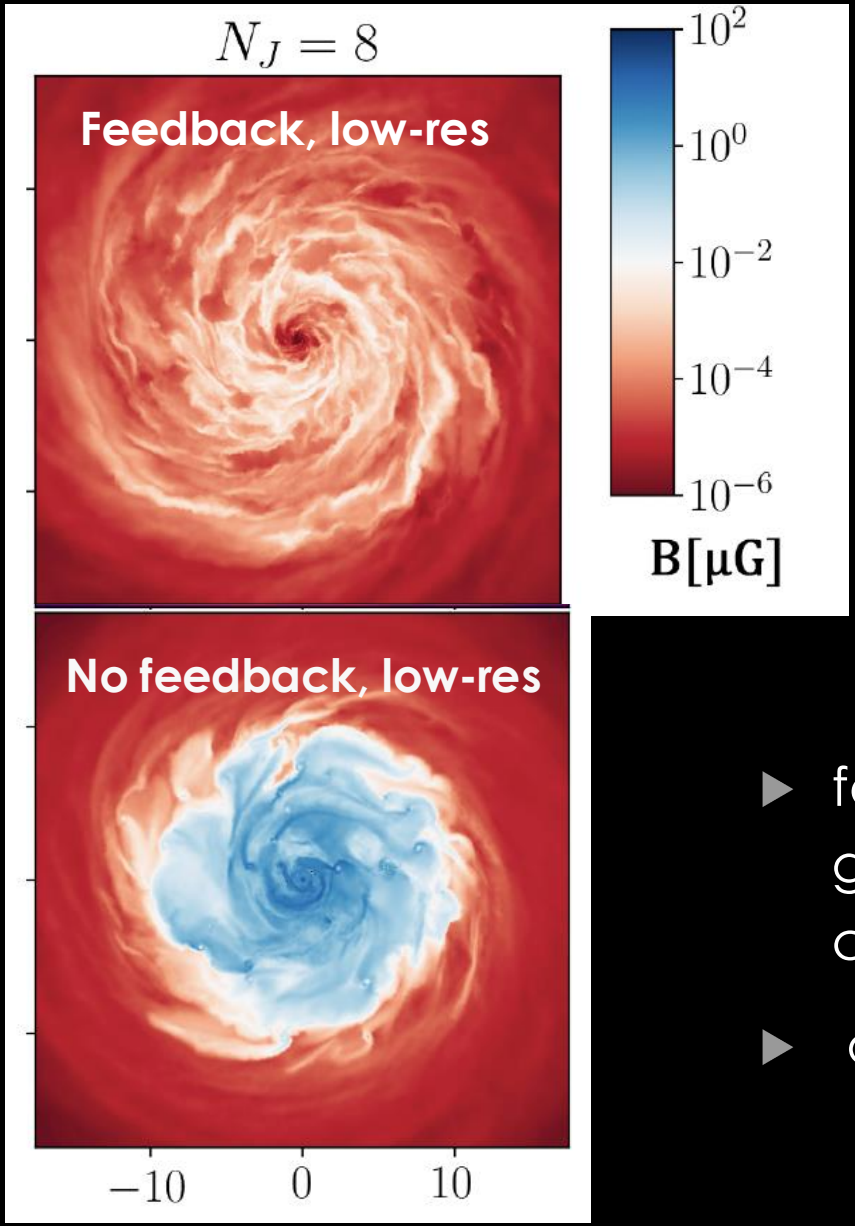


The negative & positive impact of feedback



- feedback *stablise* the disc, reduce gravitational instability (GI) and spiral arm compression -> less amplification
- advection/diffusion in the central region;

The negative & positive impact of feedback

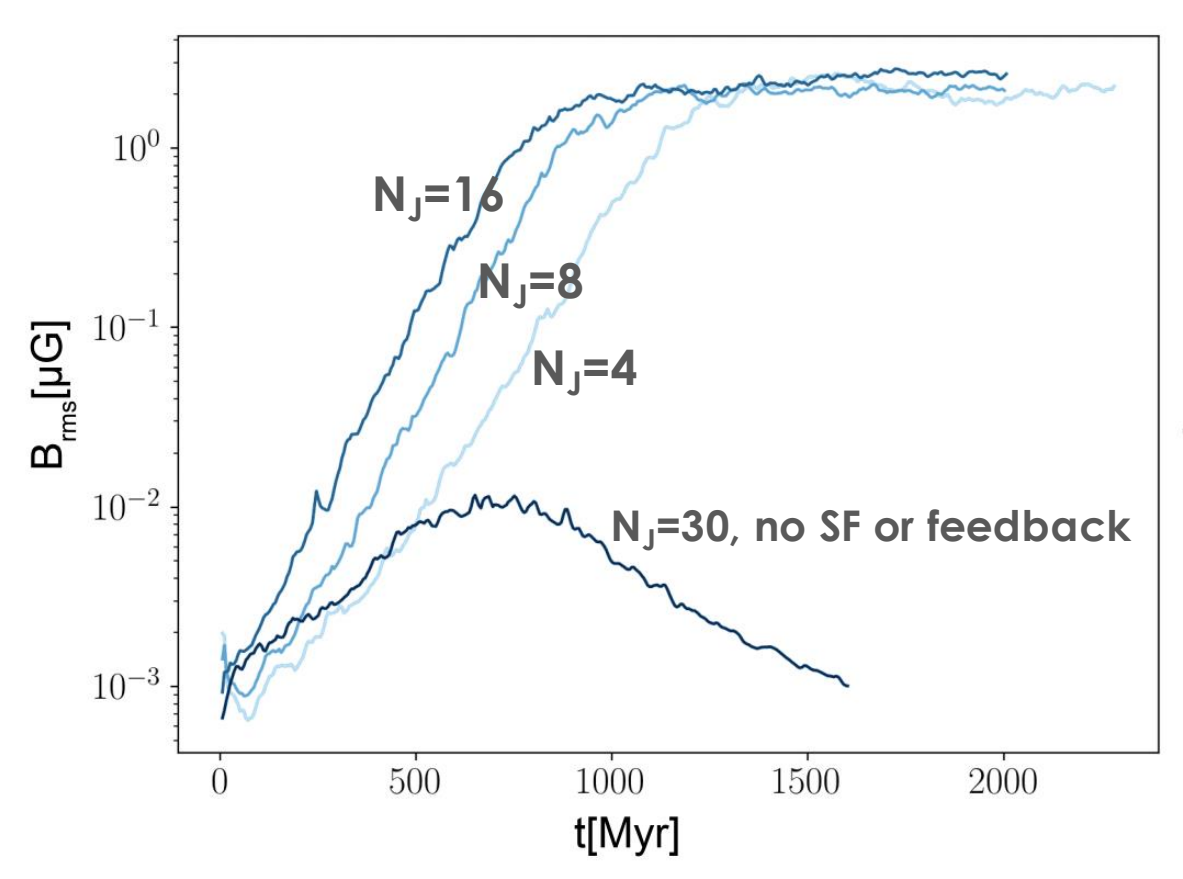




- But once one resolves the vertical structure better -> boosted amplification with feedback, esp. outer region

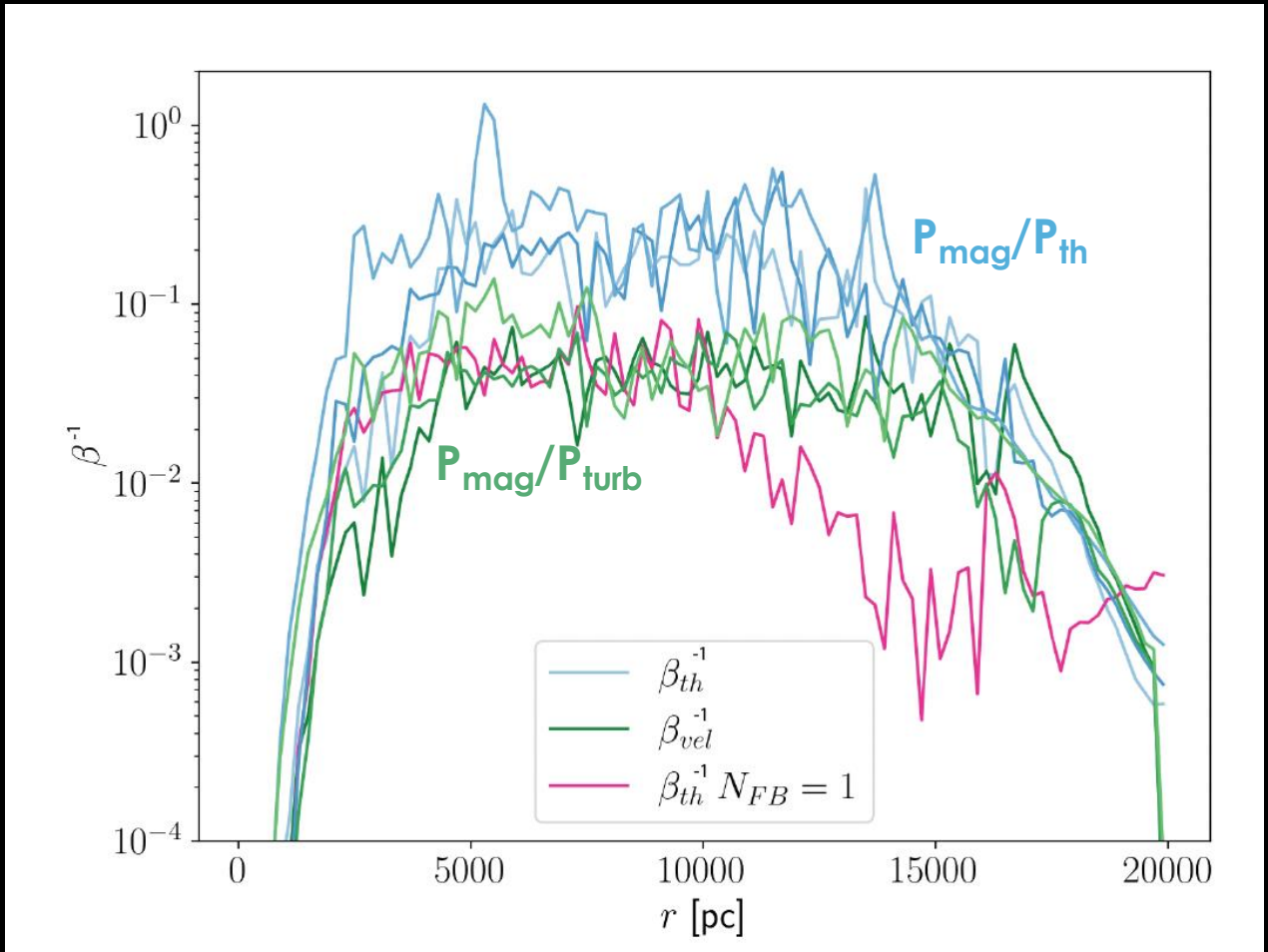
- feedback *stablise* the disc, reduce gravitational instability (GI) and spiral arm compression -> less amplification
- advection/diffusion in the central region;

Saturation levels and dependency on Jeans support



- Medium resolution with feedback runs saturate to a few μG, regardless the minimum collapsing scale;
- Larger Jeans floor -> faster amplification.
- Gravity-driven turbulence injection scale close to local Jeans scale, Federrath 2011 find resolution criteria of 30 grid cells per Jeans length to resolve dynamo.

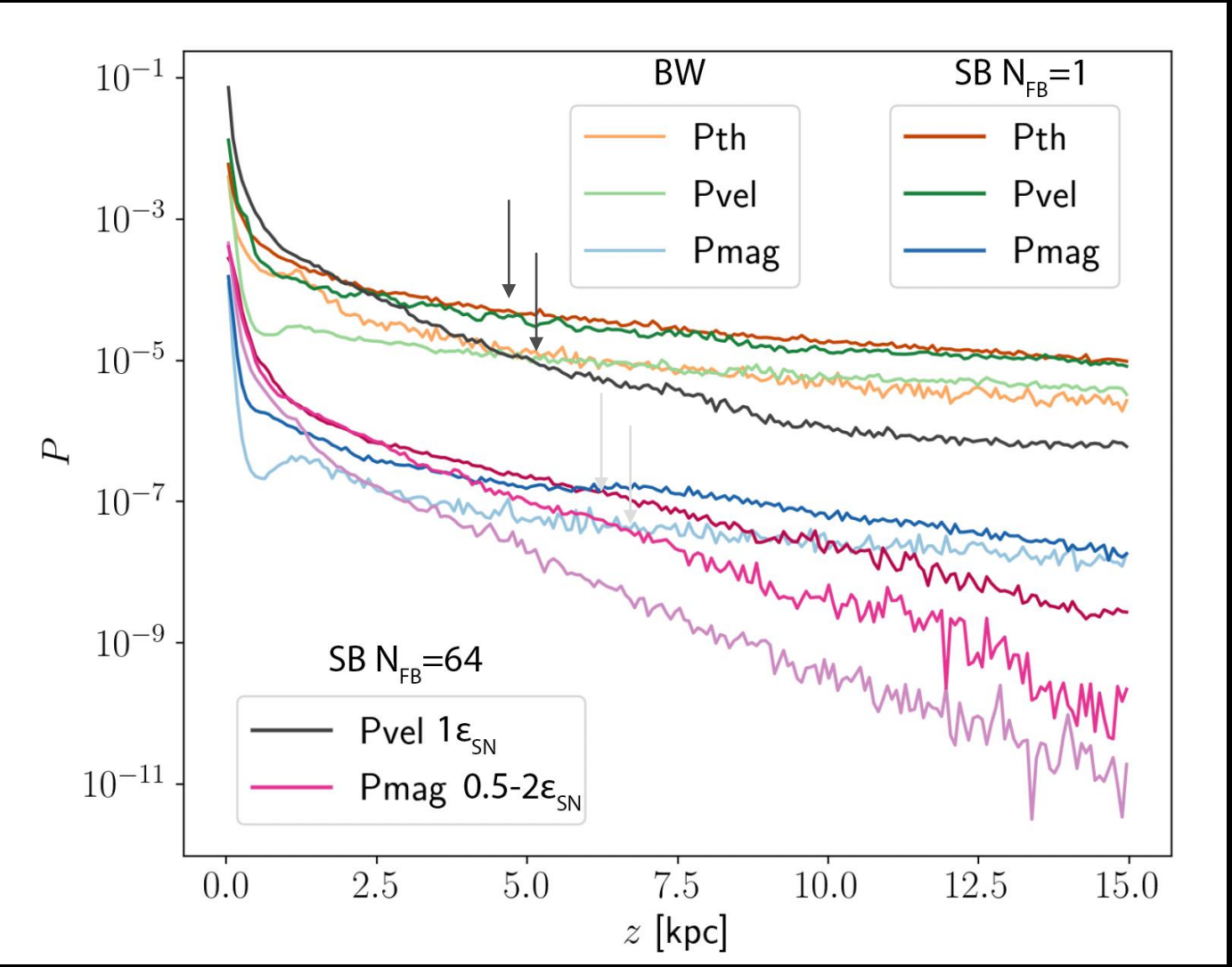
Saturation level



$\beta_x^{-1} \equiv P_{mag}/P_x$

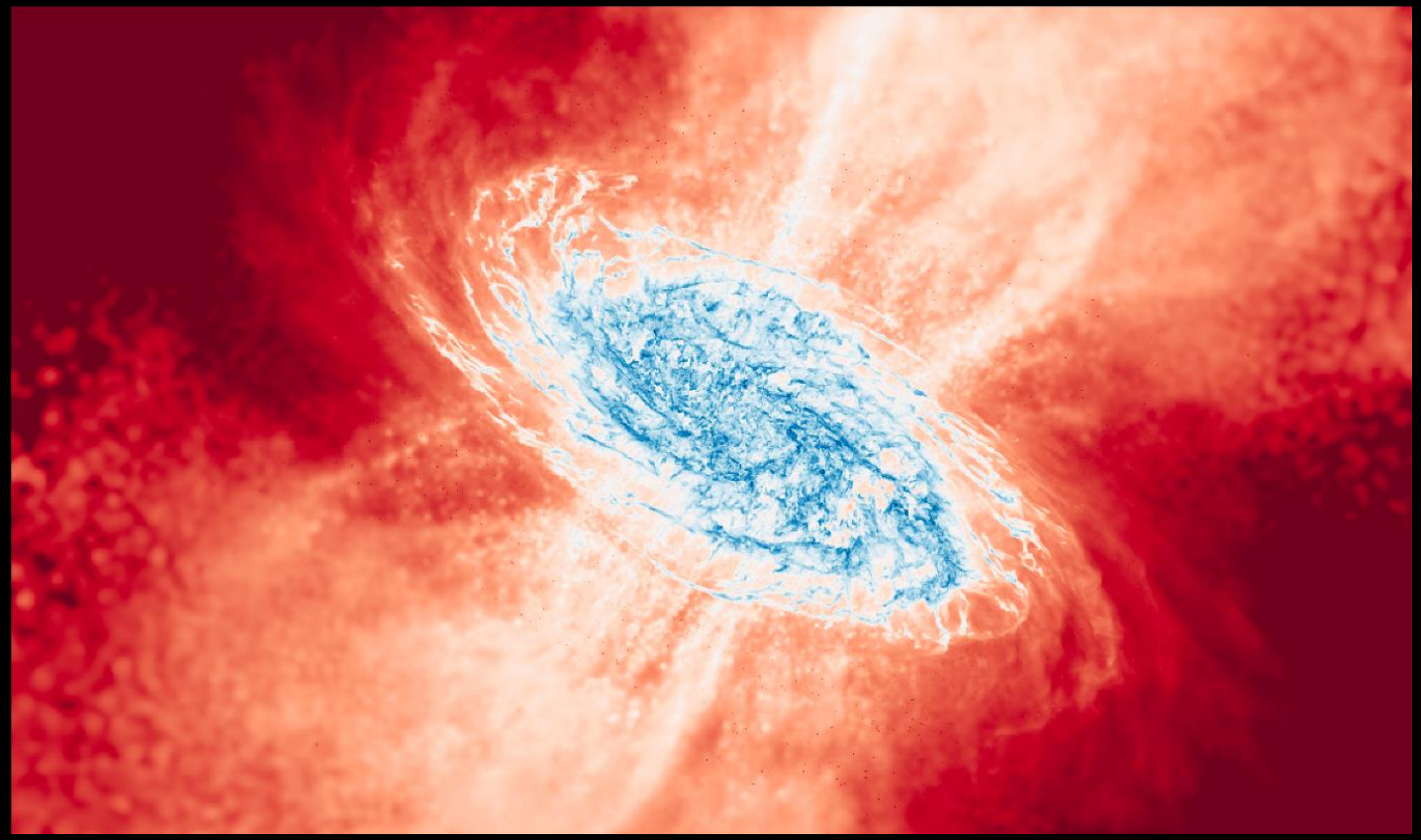
- Magnetic pressure over thermal or turbulent pressure remain flat for 4-15 kpc
- Average saturate around 10-30% thermal pressure and ~ 5% of kinetic pressure;
- B field strength similar to that of the MW

Variation on feedback models



- Correlation between P_{vel} (P_{turb}) and P_{mag}
- Saturation level within disk is largely independent of feedback
- Both model see relatively large B-field far from the disc plane driven by advection (dynamo in CGM likely not resolved)
- Higher feedback energy increase B field in the CGM as expected;

Conclusions of Wissing and Shen 2023



Main dynamo process:
Strong mean-field dynamo, consistent with the classic theory of an $\alpha\omega$ dynamo.

No Feedback:
GI sets in the disc and the dynamo is collapse driven. The smallest collapsing scale determines amplification

Feedback effect:
Destructive: Increse turbulent diffusion and eject flux into the CGM
Positive increase vertical motions and fountain motions

Resolution dependence:
High resolution dependence; but with our medium resolution the saturation level is largely independent of feedback models;

Numerical dissipation:
Prandtl number below unity
Magnetic Reynolds number in range 10-200
Around the critical value of small-scale dynamo

Initial magnetic field:
No ‘big’ effect in central disk amplification due to early flux ejection to CGM

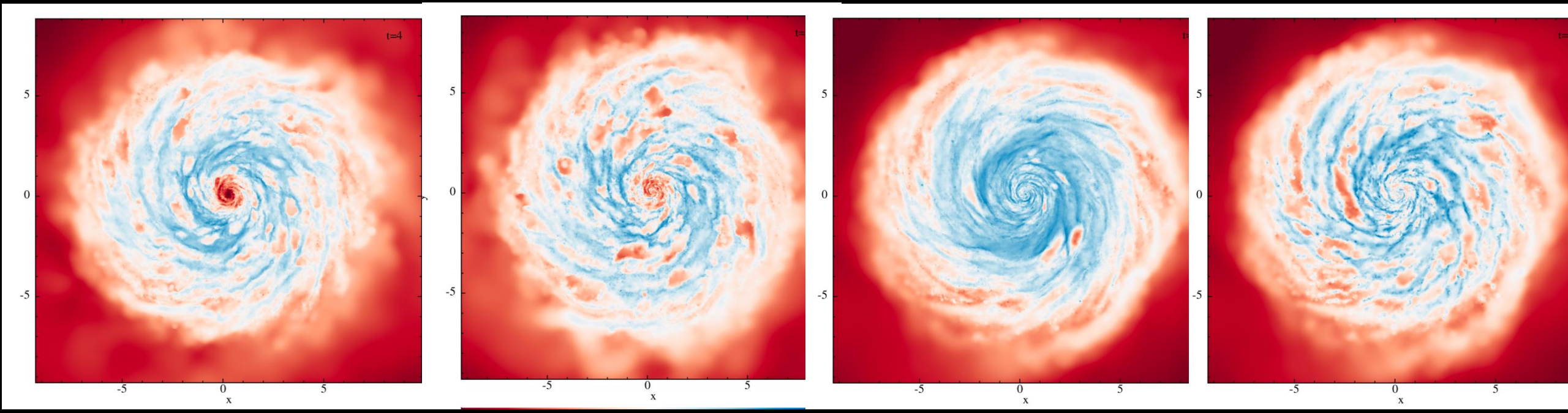
Further improvements to the SPMHD method

1. Evolve B/rho instead of B -> significantly improves magnetic field evolution when undergoing rapid expansion
2. Linear-exact gradient correction (Garcia-Senz et al. 2012)

Improving central region of galaxy

$$\frac{dB}{dt} \rightarrow \frac{d(B/\rho)}{dt}$$

--> Linear-exact gradients



GDSPH N_J=8

GDSPH N_J=8

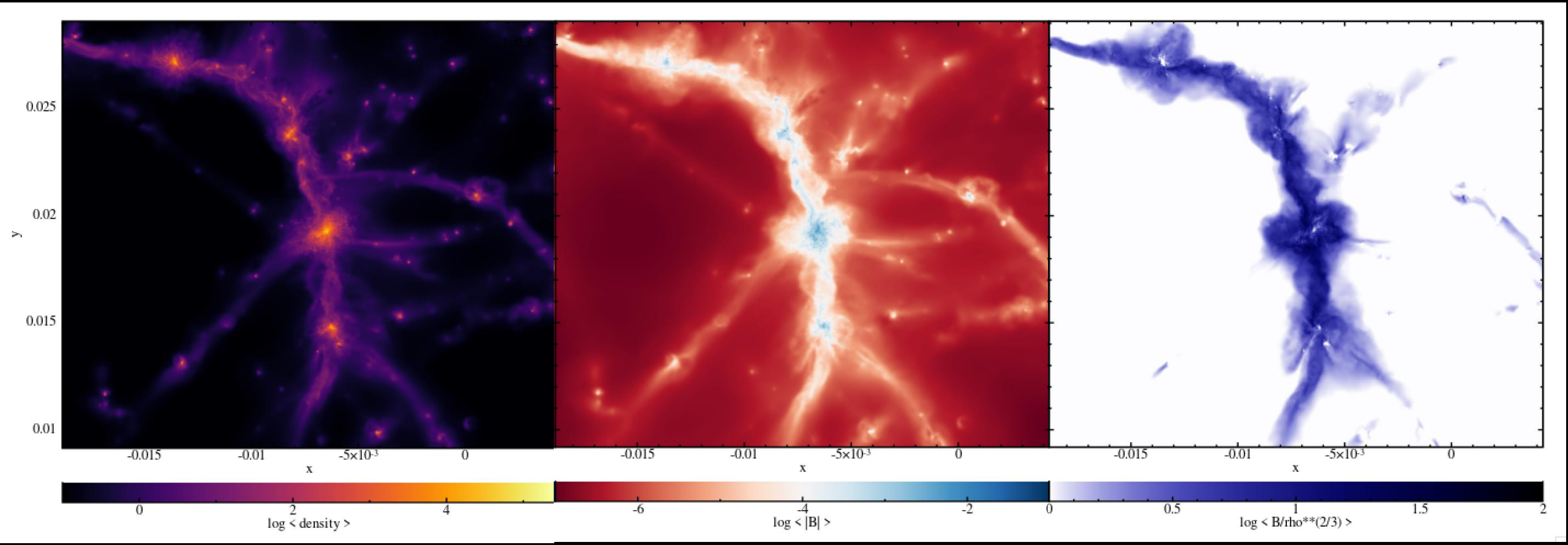
ISPH N_J=8

ISPH N_J=4

Further improvements to the SPMHD method

1. Evolve B/rho instead of B -> significantly improves magnetic field evolution when undergoing rapid expansion
2. Linear-exact gradient correction (Garcia-Senz et al. 2012)
3. Cosmological MHD implementation

Preliminary simulations of cosmological AGORA (Roca-Fabrega et al. 2024) performed to z=4



Further improvements to the SPMHD method

1. Evolve B/rho instead of B -> significantly improves magnetic field evolution when undergoing rapid expansion
2. Linear-exact gradient correction (Garcia-Senz et al. 2012)
3. Cosmological MHD implementation
4. Spin algorithm for global angular momentum conservation (non isotropic forces / linear-exact gradients) (To be submitted Wissing et al 2025)
5. Optimized smoothing kernels for SPH (To be submitted Wissing et al 2025)
6. Improved volume partitioning in SPH (To be submitted Wissing et al 2025)
7. Reconstruction scheme for dissipation (Frontiere et al. 2018)

Discrepancies between numerical codes with MHD

1. Strength of magnetic field produced
   Different simulations produce lower or higher than observed (below or above equipartition)
2. Geometry of magnetic field
   Varying degree of field reversals in disk
3. Effect on star formation
   No effect / decrease in star formation
4. What is the dominating dynamo?
   Small-scale dynamo / mean-field dynamo

Important to perform comparison project with MHD to figure out these discrepancies

AGORA!